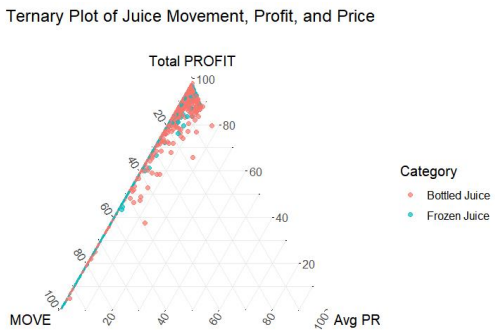
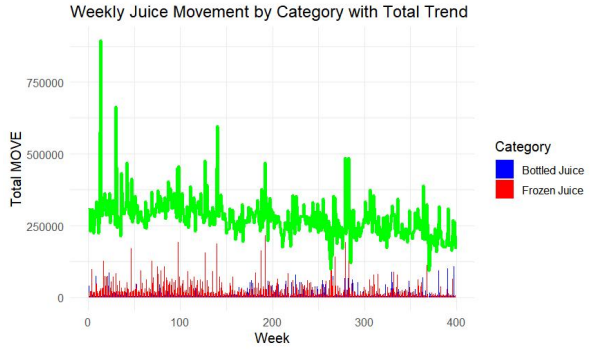
Name: Siqi Gu

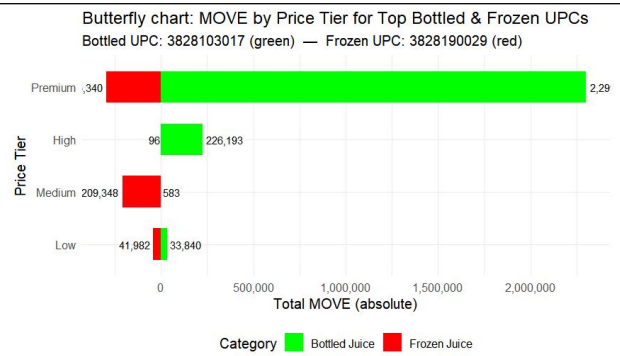
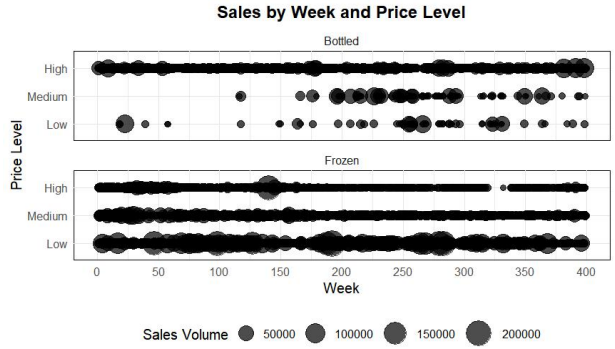
Introduction

I chose the Movement dataset from Dominick's Finer Food, focusing on Bottled Juices and Frozen Juices. I selected these categories because I am particularly interested in analyzing juice sales. My goal is to identify which types of juices have the highest sales, and what's the relationship between price,profit and move. I had originally intended to include the Refrigerated Juices dataset as well; however, unfortunately, no CSV file is available for this category.For this analysis, I kept only valid rows (OK = 1) from the Bottled and Frozen Juice datasets, and focused on the numeric columns MOVE, PRICE, and PROFIT.

Graphic 1

Design information	Graphic	Conclusion
Type of graphic: Ternary Plot Invariant: Products (Juices) Components: MOVE (Q), PROFIT (Q), PRICE (Q) Imposition: Ternary Planar variables: Position encodes ratios of MOVE/PROFIT/PRICE Retinal variables: Position Gestalt principles: Proximity, Similarity (if color used) Bertin's schemata: Associative, Ordered	 Fig 1. Ternary scatter plot (adapted from Chiera & Korolkiewicz 2017, Fig. 14).	Bottled juice perform more well across MOVE, PROFIT, and PRICE than Frozen juice. Frozen juice is high-sales but low-profit.When the price become lower, the sales become higher.

Type of graphic: Bar and Line Chart Invariant: Time (Week) Components: MOVE (Q, by category), Total MOVE (Q), Category (N) Imposition: Orthogonal (dual y-axis) Planar variables: WEEK (x), MOVE/Total MOVE (y) Retinal variables: Position, Length (bars), Color hue (category/trend) Gestalt principles: Proximity, Similarity, Good Continuation Bertin's schemata: Selective, Ordered, Size	 Fig 2. Multi-component bar chart with line	Frozen juice has higher sales. The sales decrease trend over time. About every 50 weeks, the MOVE comes to a peak.
Type of graphic: Boxplot Invariant: Price Tier Components: Total_MOVE (Q), Category (N), Price_Tier (O) Imposition: Orthogonal Planar variables: Price_Tier (x), Total_MOVE (y) Retinal variables: Position, Length (IQR/whiskers), Color hue (category) Gestalt principles: Proximity, Similarity Bertin's schemata: Associative, Ordered, Quantitative	 Fig 3. Multi-component boxplot (adapted from Chiera & Korolkiewicz 2017, Fig. 12)	Bottled juice sales concentrate in High and Premium tiers, while frozen juice dominates the Low and Medium tiers. Low tier has the most concentrated sales overall, driven almost entirely by frozen juice.The differences are extreme; the two categories have nearly opposite sales distributions across price tiers.

Type of graphic: Back-to-Bar Chart Invariant: Price Tier Components: Sales Volume (Q), Category (N), Price_Tier (O) Imposition: Symmetrical orthogonal Planar variables: Price_Tier (y), Sales Volume (x) Retinal variables: Position, Length, Color hue, Direction (L/R) Gestalt principles: Proximity, Similarity, Symmetry Bertin's schemata: Associative, Selective, Ordered	 Fig 4. Butterfly chart (adapted from Chiera & Korolkiewicz 2017, Fig. 13)	Bottle juice dominates sales when the price is high and premium. Frozen juice dominates sales when the price is low and medium. Consumers show different price sensitivity between categories.
Type of graphic: Scatter Plot Invariant: Time (Week) Components: Sales Volume (Q), WEEK (Q), Price Level (N/O) Imposition: Orthogonal Planar variables: WEEK (x), Sales Volume (y) Retinal variables: Position, Color hue (price level) Gestalt principles: Similarity, Proximity Bertin's schemata: Associative, Selective, Ordered	 Fig 5. Scatter plot (adapted from Chiera & Korolkiewicz 2017, Fig. 11)	Sales trends over time vary significantly by price level. Peak sales weeks are identifiable, often correlating with seasonal events or promotions.Peak sales weeks are identifiable, often correlating with seasonal events or promotions. Higher price levels show more sporadic sales patterns.

Type of graphic: Bubble Chart Invariant: Products/Price Levels Components: PRICE (Q), avg_profit (Q), total_sales (Q), price_level (N/O) Imposition: Orthogonal Planar variables: PRICE (x), avg_profit (y) Retinal variables: Position, Size (total_sales), Color hue (price_level) Gestalt principles: Similarity, Proximity Bertin's schemata: Associative, Selective, Ordered, Size	 Fig 6. Bubble chart (adapted from Chiera & Korolkiewicz 2017, Fig. 10)	Average profit varies significantly across price levels. The price levels that generate the highest total sales are the medium and low tiers. Higher-priced juices are more profitable in terms of unit profit but not necessarily in terms of total profit.

References

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2.Dominick's Dataset – Kilts Center | Chicago Booth